

Comparison of Proximal First-Order Primal and Primal-Dual algorithms via Performance Estimation

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Goal: Compute contraction factors of optimization methods

- Automatic computation of **worst-case performance** of a method
- Better understanding, **tuning algorithm** step-size
- Selection of **fastest algorithm**

Optimization problems: $\min_{x \in \mathcal{H}} f(x) + g(x)$ $\min_{x \in \mathcal{H}} f(x) + h(Mx)$

Optimization methods: • Gradient method • Chambolle-Pock
• Forward-backward • Condat-Vũ
• Douglas-Rachford

Performance measure: Contraction factor r :

$$\|\Phi_{f,g}x - \Phi_{f,g}y\| \leq r\|x - y\|$$

Methodology: Performance Estimation Problem

- Compute the **worst-case performance** as an optimization problem

$$\begin{aligned} \text{(PEP)} \quad & \max_{x,y,f,g} \|\Phi_{f,g}x - \Phi_{f,g}y\|^2 \\ \text{s.t.} \quad & f \in \mathcal{F}_{\rho,\alpha^{-1}}, g \in \mathcal{F}_{\mu,\beta^{-1}} \\ & \|x - y\|^2 \leq 1, \end{aligned}$$

$\mathcal{F}_{\mu,L}$: class of functions with L -Lipschitz gradient and μ -strongly convex

- **Exact convex reformulation** efficiently solvable
- Solution of (PEP) is the **largest contraction factor** for a specific algorithmic strategy Φ and a specific function class
- Available in Python library (**PEPit**) and Matlab toolbox (**PESTO**)

Application 2: Primal-dual methods

- Chambolle-Pock: $\begin{cases} x_+ = \text{prox}_{\tau f}(x - \tau M^T u) \\ u_+ = \text{prox}_{\sigma h^*}(u + \sigma M(2x_+ - x)) \end{cases}$
- Condat-Vũ: $\begin{cases} x_+ = x - \tau \nabla f(x) - \tau M^T u \\ u_+ = \text{prox}_{\sigma h^*}(u + \sigma M(2x_+ - x)) \end{cases}$

Primal- dual contraction factors

$$\|\phi_{f,g}(x, u) - \phi_{f,g}(x', u')\| \leq r\|(x, u) - (x', u')\|$$

on problem $\min_{x \in \mathcal{H}} f(x) + h(Mx)$ where $f \in \mathcal{F}_{\rho,\alpha^{-1}}, h \in \mathcal{F}_{\delta,\beta^{-1}}$ and $\|M\| \leq 1$

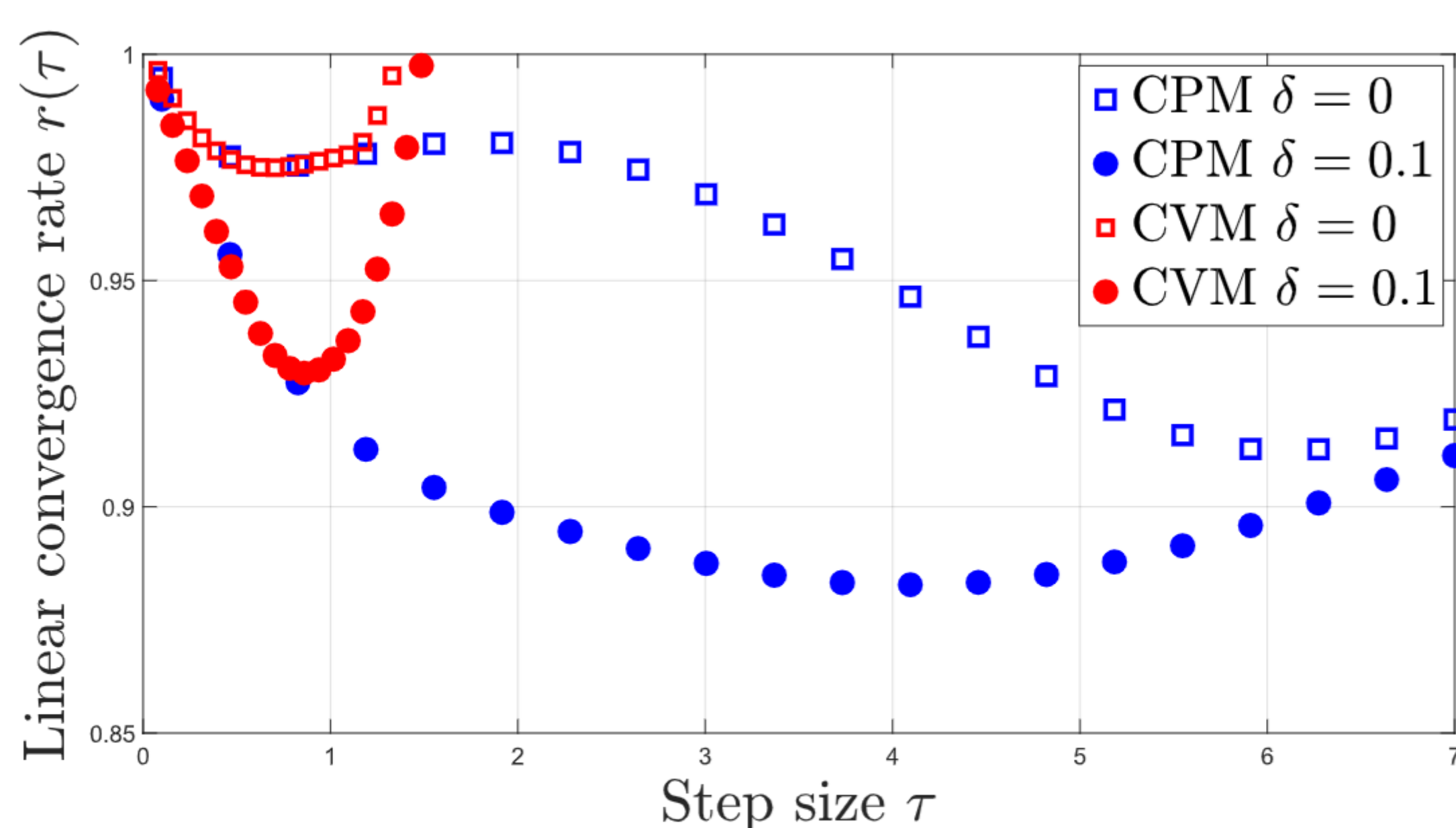
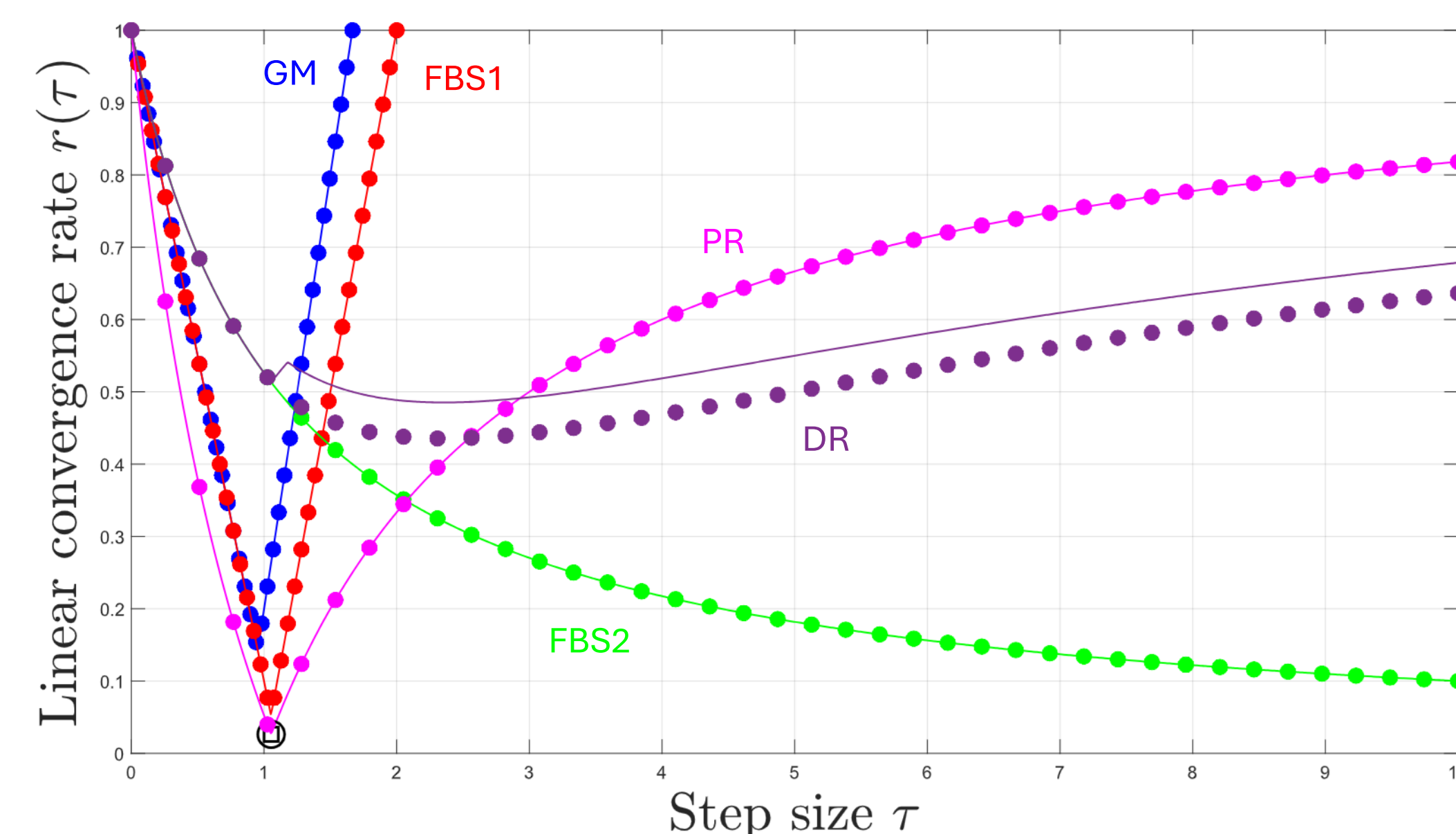


Fig. 5: Contraction factors of CPM (blue) and CVM (red) obtained by PEP for $\alpha = 1, \beta = 5, \rho = 0.1, \delta = 0$ (squares) and $\delta = 0.1$ (dots) and $\|M\| \leq 1$.

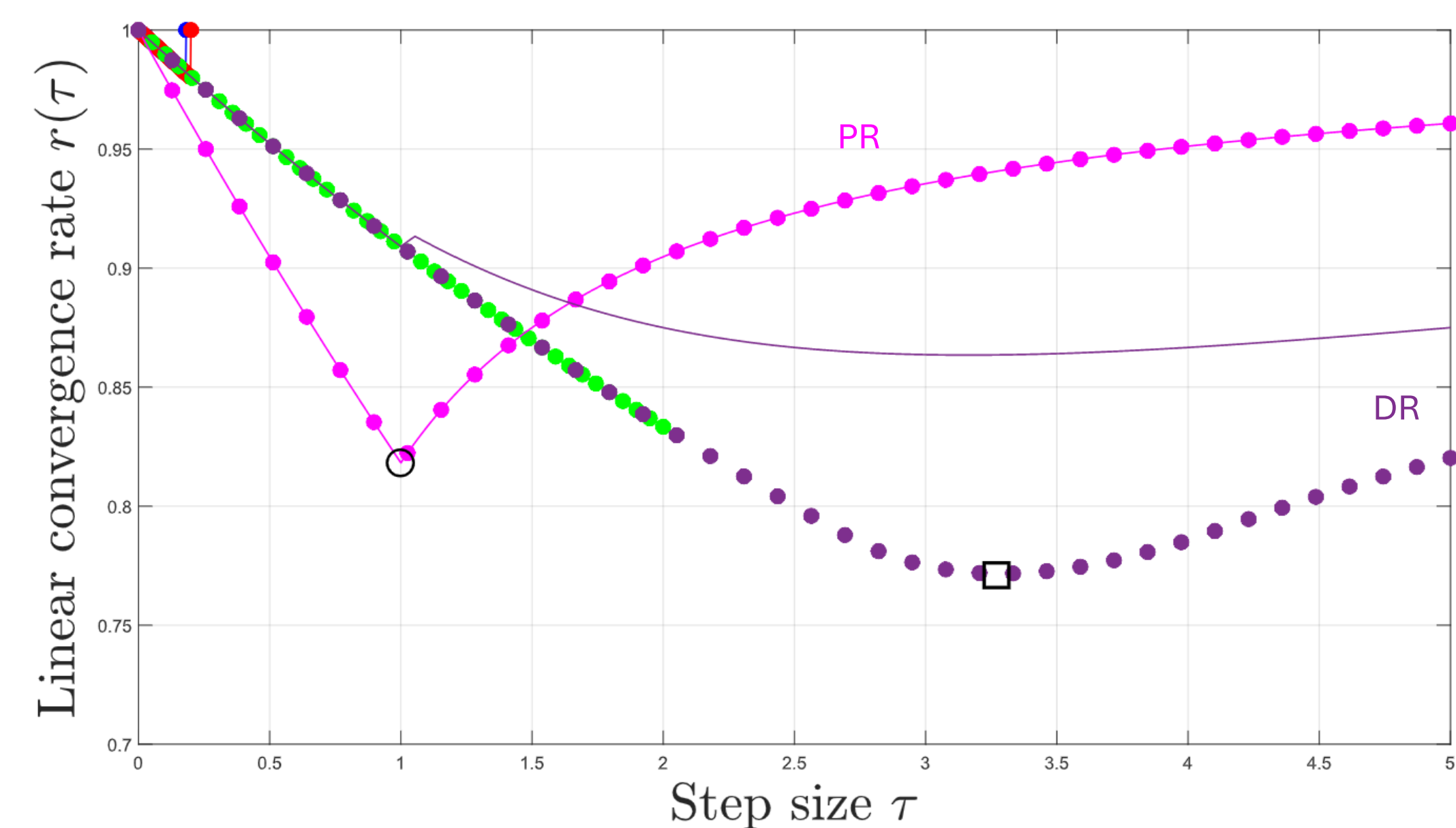
Application 1: Proximal methods

- Gradient method: $\Phi_{f,g}x = x - \tau(\nabla f(x) + \nabla g(x))$
- Forward-backward 1: $\Phi_{f,g}x = \text{prox}_{\tau g}(x - \tau \nabla f(x))$
- Forward-backward 2: $\Phi_{f,g}x = \text{prox}_{\tau f}(x - \tau \nabla g(x))$
- Peaceman-Rachford: $\begin{cases} y = \text{prox}_{\tau f}(x) \\ \Phi_{f,g}x = 2\text{prox}_{\tau g}(2y - x) - 2y + x \end{cases}$
- Douglas-Rachford: $\begin{cases} y = \text{prox}_{\tau f}(x) \\ \Phi_{f,g}x = \text{prox}_{\tau g}(2y - x) - y + x \end{cases}$

Contraction factors from [Briceño-Arias, Pustelnik 2023] (lines) and computed by PEP (dots)



(a) $\alpha = 1, \beta = 5, \rho = 0.9, \mu = 0$.



(b) $\alpha = 0.1, \beta = 1, \rho = 0.1, \mu = 0$.

Observations:

- Results from [Briceño-Arias, Pustelnik 2023] **were tight** except for Douglas-Rachford
- Worst-case contraction attained by **simple univariate quadratic** functions
- Exact analysis allows **selecting optimal method**

References:

- Bousselmi, N., Hendrickx, J. M., & Glineur, F. (2023). Interpolation conditions for linear operators and applications to performance estimation problems. *SIAM Journal on Optimization*, *accepted*.
- Briceño-Arias, L. M., & Pustelnik, N. (2023). Theoretical and numerical comparison of first order algorithms for cocoercive equations and smooth convex optimization. *Signal Processing*, 206, 108900.

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